

5	(a) <u>total/sum</u> of electromotive forces or e.m.f.s = <u>total/sum</u> of potential differences or p.d.s around a loop/(closed) circuit	M1 A1	[2]
(b)	(i) (current in battery =) current in A + current in B or $I_A + I_B$ $(I =) 0.14 + 0.26 = 0.40 \text{ A}$	C1 A1	[2]
	(ii) $E = V + Ir$ $6.8 = 6.0 + 0.40r$ or $6.8 = 0.40(15 + r)$ $r = 2.0 \Omega$	C1 A1	[2]
	(iii) $R = V/I$ ratio ($= R_A/R_B$) = $(6.0/0.14)/(6.0/0.26)$ = $42.9/23.1$ or $0.26/0.14$ = $1.9 (1.86)$	C1 A1	[2]
	(iv) 1. $P = EI$ or VI or $P = I^2R$ or $P = V^2/R$ $= 6.8 \times 0.40$ $= 0.40^2 \times 17$ $= 6.8^2/17$ $= 2.7 \text{ W} (2.72 \text{ W})$	C1 A1	[2]
	2. output power = VI = $6.0 \times 0.40 (= 2.40 \text{ W})$ efficiency = $(6.0 \times 0.40)/(6.8 \times 0.40) = 2.40/2.72$ = 0.88 or 88% (allow 0.89 or 89%)	C1 A1	[2]

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